- Curse of dimensionality: If we use more features, we need to go further from center to get significant portion of data.
- Otsu's Method: Mixed Var $= \frac{(\omega_L \times \sigma_L^2 + \omega_R \times \sigma_R^2 \times 10)}{10}$ (Noise ↓)
- Mixture model: Each intent class from diff^t ND.
- Gradient Descent / Convex Optimization (Loss function)
  - gradient > 0 (function ↑, move left, ↓ parameters)
  - gradient ≤ 0 (" ↓, " right, ↑ ")
  - Hyperparameter Tuning → Grid search / → Adaptive grid search
- Fast Rejection Cascade (Feature Selection)

Cost function = Obj func + Regularization (↓ K-means.)

Measure$_{node}$ = Size$_{node}$ + α (purity) ↓ Better

fraction of majority class

① Find best threshold across features meeting that condition by their majority class ② Classify all samples correctly classified ③ Remove those samples and repeat on remaining data ④ Stop when 95% of data is correctly classified

Size$_{node}$ × Purity − α (impurity)

Rewards size & purity (Obj) — Penalises it (Reg)

(Stay on road to reach dist but not at 1 Mph)

- Gaussian = $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]}$ Shifts to right by μ → spreads out (Measure of dispersion)

(Kurtosis - Gaussian, Skewness - Shifts the model is)

- Blind Signal Decomposition

→ Randomly choose ω ⇒ compute O/P signal: $y = \omega^T x$

(Noise: Gaussian) (real data)

→ Measure non-gaussianity of y (Kurtosis)

→ Update ω using Gradient Ascent − ↑ Non-Gauss.

(Repeat until convergence (y becomes independent source)

Kurtosis → how peaked / flat distribution is

Excess Kurtosis → how "peaked" or flat a distribution is compared to normal distr. (≈0 Noise)

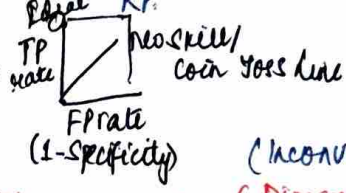
- Haversine distance / Great Circle Dist → dist. b/w 2 points on Earth.

- Ensemble Classifier: Train multiple classifiers on same dataset. Each classifier makes prediction. ①

Combine → Votes: majority vote: Classification; Averaging / weighting → Regression.

(↓ var: prevents overfitting, ↓ bias → improves accuracy)

- Data Reduction: Linear interpolation.
- Comodality: If 2 things happen at same event, we can see only 1 event.
- False Alarm Rate = FP / FP + TN
- True Positive Rate (Recall / Sensitivity) = TP / TP + FN (How many true reported out of all)
- False Negative Rate (Miss) = FN / FN + TP
- Precision = TP / TP + FP (Positive Predictive Value)
- True Negative Rate (Specificity) = TN / TN + FP
- F1 score = $\frac{2 * R * P}{R P}$
- ROC Curve



AUC → how well it classifies

- Norm Distance Metric (Dimensionality Reduction) (Inconvenience, dissimilarity)
- Norm Theory → Positivity (No -ve dist. b/w x, y); → Symmetry (dist. from x to y = y to x); → Triangle Inequality (diff. in amount of ingr. of recipes) $A \triangle B$ C, $AC \leq AB + BC$
- L_1 Norm / Manhattan / Taxi-cab → diamond shaped (Round shaped) → opposite effect as Euclidean dist (L_2 Norm)
- L_0 Norm (special case) → measures sparsity → how many attributes features are active (Feature Selection) (No. of diff ingredients in recipe)

adds one rule to cascade

- L_∞ Norm (Supremum) → ~~Special case~~ Largest absolute value among components of vector.
- Minkowski Norm / L_p ⇒ $L_p \triangleq \left(\sum_{k=1}^{n} |x_k - y_k|^p\right)^{1/p}$

Mahalnobis distance − Elliptical clusters.

- Maths: − Conversion to Spherical dist. → Euclidean.
- Prob. Density: How many contour intervals you cross to get to a point from centre
- Analogy (Centre of gaussian − peak → as we go out height ↓) → so it is generalized z-score in multi-dimensional space.

Hamming distance – special case of L0 Norm. (counts non-zero diff.)

applies it specifically to binary vectors. ($f_{01} + f_{10}$)

Simple Matching Coeff (Normalization) :→ $\frac{f_{00} + f_{11}}{All}$

Jaccard Similarity Coeff :→ $f_{11} / f_{10} + f_{01} + f_{11}$ (matching 2 present)

↳ form clusters of customers

Cosine Similarity → do concept mapping first

$\cos\theta = 1$ same dir, -1 opp dir

↳ text comparison.

↳ useful for any vectors in any no. of dimensions

XDNA analysis

Mutual Info Amount of info obtained about one variable by observing second variable

$I(s,t) = \sum_{s \in S} \sum_{t \in T} p(s,t) \log_2 \left(\frac{p(s,t)}{p(s)\,p(t)} \right)$ → independent = 1, $I = 0$

Z-score Outlier Removal (Normalise data) $CC(s,t) = \frac{1}{N-1} \sum \frac{(s-\mu_s)(t-\mu_t)}{\sigma_s}$

How far is SD of each of these dist. from point. (1 dist – classify in that) (same for MD)

(Diff^t distances give diff^t clusters) – K-Means with

Manhattan dist. ~~gives~~ tends towards unit circle.

↑ coefficient: When s is above mean, t tends to be above mean too

Cross-Correlation Analysis (Feature Selection)

↳ Feature selection & rejection → improves quality of later

• If feature has high correlation with particular class can be used for classification (identifies redundancy)

PCA (correlation b/w variables ↑ – singular component)

→ guides DR but not DR.

Histogram Intersection: Min^m of 2 hist. for each bin, then add

abs-diff → Standard His. Difference.

Earth Mover's distance: how much, how far to move (diff in size)

Bhattacharya Coefficient: Similarity b/w 2 PDs.

↳ How hard it is to separate one PD from another

↳ $BC(p,q) = \sum_{k=1}^{N} \sqrt{p(k)q(k)}$ (=1 → same.)

Levenshtein dist. → Insertion, deletion, substitution (spell check) (oops dist – keyboard)

Full K-Means

① for large no. of iterations

a. Randomly select (K) points

b. Repeat

→ Assign all pts to closest center

→ for each new cluster formed, find new prototype center for cluster

→ Stop when prototypes donot move.

c. Evaluate resulting cluster cohesion for each set of seed points and remember best clustering and prototypes ② Use best set of prototypes

SSE | knee point (inflection pt) (↑K → ↓ SSE)

$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} dist^2(m_i, x)$ (cluster i, rep. pt, actual pt)

Stirling number $= \left\{ {n \atop k} \right\}$ → No. of possible ways to assign pts to cluster

↳ Helps in understanding initialization space.

→ sub sample (another clus. algo)

→ select ↑ K initial centroids (select among these – most widely sep.)

→ Multiple runs: Run with lowest SSE.

Failure (K-Means) → diff^t sizes of clusters

→ diff^t densities (↑ Noise)

→ Concave clusters → bad segmentation (Non-globular)

→ Outliers

density based clusters

cluster within cluster → DBSCAN

Ignore pts outside of some range & repeat as needed)

(Biggest cluster → most datapts / SSE / Var / combined)

Bisecting K-Means clustering

• Initialize all data to one cluster

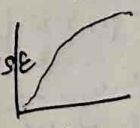
• Repeat (until K clusters)

→ select biggest cluster

↳ delete from list.

↳ bisect it using 2-means

↳ Add 2 clusters to list



Pre-processing: Normalize data, outliers ×

Post-processing: Eliminate small clusters (may represent outliers)

↳ split loose clusters (clust. with high SSE)

↳ merge clusters that are close & have relatively low SSE

Apriori Principle (Recommender Algo) Coupons

→ If itemset is frequent, then all of its subsets are also frequent

set → joint condition

↳ subsets meet this

we list all 2-item combinations, count how many transactions contain both items & keep only those above threshold